```
PS D:\Tasks\Python\KnowledgeEngineering\Task1> python -u "d:\Tasks\Python\KnowledgeEngineering\Task1\F1-curve.py"
[LibLinear]iter  1 act 2.250e+04 pre 2.000e+04 delta 1.197e-02 f 2.268e+05 |g| 3.817e+06 CG   1
iter  2 act 1.974e+03 pre 1.781e+03 delta 1.197e-02 f 2.043e+05 |g| 8.182e+05 CG   1
cg reaches trust region boundary
iter  3 act 6.641e+02 pre 6.614e+02 delta 4.790e-02 f 2.023e+05 |g| 1.356e+05 CG   2
cg reaches trust region boundary
iter  4 act 2.235e+03 pre 2.238e+03 delta 1.916e-01 f 2.017e+05 |g| 5.224e+04 CG   2
cg reaches trust region boundary
iter  5 act 8.727e+03 pre 8.682e+03 delta 7.663e-01 f 1.994e+05 |g| 1.930e+05 CG   2
cg reaches trust region boundary
iter  6 act 2.152e+04 pre 2.084e+04 delta 1.199e+00 f 1.907e+05 |g| 5.129e+04 CG   2
cg reaches trust region boundary
iter  7 act 1.326e+04 pre 1.203e+04 delta 1.565e+00 f 1.692e+05 |g| 1.280e+05 CG   3
cg reaches trust region boundary
iter  8 act 8.765e+03 pre 9.279e+03 delta 1.643e+00 f 1.559e+05 |g| 9.213e+04 CG   3
cg reaches trust region boundary
iter  9 act 9.028e+03 pre 8.399e+03 delta 2.476e+00 f 1.472e+05 |g| 5.240e+04 CG   3
iter 10 act 1.148e+01 pre 1.140e+01 delta 2.476e+00 f 1.381e+05 |g| 3.414e+04 CG   1
cg reaches trust region boundary
iter 11 act 4.779e+03 pre 4.325e+03 delta 3.159e+00 f 1.381e+05 |g| 3.244e+03 CG   3
cg reaches trust region boundary
iter 12 act 2.136e+03 pre 1.961e+03 delta 3.472e+00 f 1.334e+05 |g| 5.229e+03 CG   5
iter 13 act 1.731e+00 pre 1.726e+00 delta 3.472e+00 f 1.312e+05 |g| 1.238e+04 CG   1
cg reaches trust region boundary
iter 14 act 1.017e+03 pre 9.970e+02 delta 3.686e+00 f 1.312e+05 |g| 9.108e+02 CG   8
iter 15 act 4.454e-01 pre 4.448e-01 delta 3.686e+00 f 1.302e+05 |g| 6.250e+03 CG   1
cg reaches trust region boundary
```

```
Epoch 3, change: 0.21559892
Epoch 4, change: 0.16264426
Epoch 5, change: 0.13230134
Epoch 6, change: 0.11500933
Epoch 7, change: 0.10123396
Epoch 8, change: 0.09097584
Epoch 9, change: 0.08061944
Epoch 10, change: 0.07232753
Epoch 11, change: 0.06576585
Epoch 12, change: 0.05986116
Epoch 13, change: 0.05539335
Epoch 14, change: 0.05101094
Epoch 15, change: 0.04754384
Epoch 16, change: 0.04444247
Epoch 17, change: 0.04155099
Epoch 18, change: 0.03915645
Epoch 19, change: 0.03688893
Epoch 20, change: 0.03473441
Epoch 21, change: 0.03300229
Epoch 22, change: 0.03131427
Epoch 23, change: 0.02980657
Epoch 24, change: 0.02849466
Epoch 25, change: 0.02714118
Epoch 26, change: 0.02593871
Epoch 27, change: 0.02482845
```

```
Epoch 1/10
5973/5973 ━━━━━━━━━━━━━━━━━━━━ 4s 568us/step - accuracy: 0.9643 - loss: 0.1387
Epoch 2/10
5973/5973 ━━━━━━━━━━━━━━━━━━━━ 3s 566us/step - accuracy: 0.9679 - loss: 0.1198
Epoch 3/10
5973/5973 ━━━━━━━━━━━━━━━━━━━━ 3s 562us/step - accuracy: 0.9671 - loss: 0.1196
Epoch 4/10
5973/5973 ━━━━━━━━━━━━━━━━━━━━ 3s 574us/step - accuracy: 0.9673 - loss: 0.1173
Epoch 5/10
5973/5973 ━━━━━━━━━━━━━━━━━━━━ 4s 588us/step - accuracy: 0.9675 - loss: 0.1161
Epoch 6/10
5973/5973 ━━━━━━━━━━━━━━━━━━━━ 4s 585us/step - accuracy: 0.9673 - loss: 0.1162
Epoch 7/10
5973/5973 ━━━━━━━━━━━━━━━━━━━━ 3s 565us/step - accuracy: 0.9672 - loss: 0.1157
Epoch 8/10
5973/5973 ━━━━━━━━━━━━━━━━━━━━ 3s 576us/step - accuracy: 0.9681 - loss: 0.1132
Epoch 9/10
3199/3199 ━━━━━━━━━━━━━━━━━━━━ 1s 412us/step - accuracy: 0.9644 - loss: 0.1240
验证集损失: 0.13002324104309082, 验证集准确率: 0.9622040390968323
3199/3199 ━━━━━━━━━━━━━━━━━━━━ 1s 352us/step
Precision-Recall AUC: 0.20
3199/3199 ━━━━━━━━━━━━━━━━━━━━ 1s 364us/step
ROC AUC: 0.83
```